Smart Rental Intelligence Platform

Project Overview & Technology Summary

A dealer-personalized, AI-assisted equipment rental tracking platform for construction & infrastructure fleets, built for the Caterpillar-style hackathon brief.

1. What This Project Is

Construction and infrastructure companies rent heavy equipment (excavators, cranes, bulldozers, backhoes, graders, loaders, forklifts, compactors) but typically track rentals via spreadsheets, WhatsApp, and paper — leading to lost equipment, missed returns, misallocation, and no real analytics. This platform replaces that with a single, real-time dashboard: live equipment status and location, QR-based check-in/check-out, automatic overdue alerts, rule-based demand forecasting and anomaly detection, predictive maintenance scoring, and a conversational AI assistant that can answer questions about the fleet in plain English.

The platform is scoped to **one dealer managing a personalized fleet of 10 vehicles**, all operating within India, seeded from a real synthetic telemetry dataset and kept live by an external Python client that posts fresh readings to the API roughly every 10 seconds — standing in for a real IoT feed.

2. Technology Stack

Layer	Technology	Role
Frontend	React 18 + TypeScript, Vite	Single-page dealer dashboard application
Styling	Tailwind CSS	Utility-first styling, responsive layout
Charts	Recharts	Utilization, fuel usage, revenue, and fleet-status visualizations
Maps	Leaflet + react-leaflet	Live equipment map with auto-fit bounds and geofence-compliance color coding
Backend API	FastAPI (Python)	REST API, request validation via Pydantic, async lifespan tasks
Database	SQLAlchemy ORM + SQLite	Normalized relational schema, designed to be Postgres-compatible for production
AI orchestration	LangGraph	Multi-agent state graph: an LLM router node dispatches to 5 specialist agents
AI framework	LangChain (langchain-core, langchain-groq)	Tool-calling, structured output, ReAct agent construction
LLM	Groq — Llama 3.3 70B Versatile	Powers the routing and all 5 specialist agents
QR codes	qrcode (Python)	Generates a QR image per rental for check-out
Live data client	Python + requests	Standalone script simulating a real-time IoT telemetry feed via the REST API

3. Architecture

The backend follows a strict separation between **routes**, **services**, and **agents**, so every rule (anomaly detection, forecasting, alerts, maintenance scoring) lives in a single reusable service function — callable from a REST endpoint, the background safety-net job, *or* an AI agent tool, with no duplicated logic.

- **Database schema:** Dealers, Customers, Equipment, Sites, Rentals, Payments, Telemetry, UsageLogs, Notifications, Forecasts, Agreements, and **EquipmentTenure** (each vehicle's designated home coordinates and how long it's been deployed there).
- **Service layer:** anomaly_service, forecast_service, alert_service, maintenance_service, telemetry_ingest — pure, testable business logic with no route-handling code mixed in.
- **Agent layer:** a tools.py wrapping every service as a read-only LangChain tool, five specialist agents each bound to a relevant subset of tools, and one orchestrator agent that classifies a dealer's question and routes it to the right specialist.
- **Real-time data path:** an external script POSTs a live reading for every vehicle to `/api/telemetry/report` every ~10 seconds; the backend updates the vehicle and immediately re-runs overdue, anomaly, and maintenance checks.

4. Core Features

4.1 Dealer Dashboard & Fleet Overview

- At-a-glance stat cards: Total Fleet, Out on Rent, Available Now, Overdue.
- "Currently rented — live status" table: which customer has which vehicle, when it was checked out, expected return date, and a color-coded status (on-time / due tomorrow / overdue by N days).
- "Available now" quick list of every vehicle ready to rent.
- Full fleet detail table with live runtime hours, fuel usage, idle hours, engine health %, and return date per vehicle.

4.2 Live Equipment Map

- Every vehicle plotted on an interactive map, auto-fitted to bounds so the whole fleet stays visible however geographically spread out it is.
- Markers color-coded by geofence compliance: blue when a vehicle is within its designated coordinates, red when it has drifted outside them.
- For a non-compliant vehicle, a dashed line connects its live position to its designated "home" location for immediate visual context.

4.3 Check-In / Check-Out with QR Codes

- Two-step rental flow: the dealer first checks live machine availability, then creates a new customer record before a rental can be created — enforcing that order by design.
- A QR code is generated per rental and the equipment is checked out automatically the moment the QR is issued (no extra manual click needed).
- Check-in is a simulated "scan" action that closes out the rental and returns the vehicle to the available pool.

4.4 Real-Time Telemetry Feed

- A standalone Python script (`live_telemetry_client.py`) fetches the fleet and posts a fresh, realistic reading — fuel, runtime hours, idle hours, GPS — for every vehicle roughly every 10 seconds, entirely through the REST API.
- Each posted reading updates the database and immediately re-triggers overdue, anomaly, and maintenance checks, so the dashboard reflects new data without any manual refresh action.
- Occasional simulated GPS drift events let the geofence-violation alerting be demonstrated live rather than only at startup.

4.5 Geofence & Tenure Tracking

- A dedicated EquipmentTenure table stores each vehicle's precise designated coordinates (distinct from the shared site location) and how long it has been deployed there.
- The system continuously compares live GPS against these designated coordinates and flags the dealer the moment a vehicle strays outside its assigned geofence — a signal for possible unauthorized movement.

4.6 Summary Analytics

- Utilization %, total rented hours, total idle hours and idle %, fuel usage, and an estimated revenue figure.
- Usage-per-site bar chart and a fleet-utilization pie chart (rented vs. available/idle).

4.7 Overdue Alerts

- Automatic detection of rentals due back tomorrow or already overdue, surfaced as a persistent banner and in a dedicated Alerts feed.
- Re-evaluated on every telemetry update, so the alert list never goes stale.

4.8 Demand Forecasting

- Rule-based ranking of which equipment types have historically been rented most, per site — no black-box ML, every number is traceable to real rental history.
- Pre-positioning recommendations: for high-demand site/type combinations without a matching unit already on site, the system recommends a specific vehicle to move there and states when it becomes available.

4.9 Anomaly Detection

- Rule-based checks across the fleet: excessive idle hours, unassigned equipment, geofence violations, unexpected movement, fuel-usage anomalies, zero usage on an active rental, and degrading engine health.
- Every anomaly includes a severity level and a concrete recommended action.

4.10 Predictive Maintenance

- An explainable 0–100 risk score per vehicle, combining engine health, continuous-tenure duration, idle-to-runtime ratio, fuel anomalies, and current geofence compliance.
- High-risk vehicles automatically appear in the Alerts feed with a recommended inspection action.

4.11 AI Multi-Agent Assistant

A conversational "AI Assistant" tab lets the dealer ask plain-English questions. A LangGraph orchestrator classifies each question and routes it to exactly one of five specialist agents, each with its own focused toolset:

Agent	Handles
Fleet Agent	Equipment status, live location, geofence compliance, and "which vehicles are near <Indian city>" style questions.
Rental Scheduling Agent	Availability checks, rental history, and overdue returns. Advisory only — it never creates or modifies a rental itself.
Demand Forecast Agent	Ranked demand by site/type, plus concrete pre-positioning recommendations.
Smart Alert Agent	Current active alerts and on-demand fresh anomaly scans, prioritized by severity.
Predictive Maintenance Agent	Per-vehicle and fleet-wide maintenance risk scoring with cited reasons.

Questions unrelated to fleet or rental management are recognized and politely declined by a dedicated out-of-scope path, instead of being force-routed into a specialist that would otherwise fabricate an answer.

5. Data & Scope

- Fleet: 10 vehicles (a mix of excavators, cranes, bulldozers, backhoes, graders, and loaders), personalized to represent one dealer's real operation rather than a large generic dataset.
- Geography: every coordinate in the system — sites, designated locations, and live GPS — is anchored to real Indian cities (Chennai, Hyderabad, Mumbai, Pune, Delhi, Kolkata).
- Seed data: derived from a 350-row synthetic telemetry dataset, trimmed to the personalized 10-vehicle fleet; ongoing values are then kept current by the live telemetry client.

Generated as a project overview for the Smart Rental Intelligence Platform.